

EM-II
Practice Question Sheet #4
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Topics: Gram-Schmidt Method, Differential Equation & PDE.

1. Let $W = \text{span}\{(1, 2, -1)\} \subset R^3$, find orthogonal projection of $v = (-10, 2, 8)$ onto W .
2. Let $S = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\} \subset R^3$. Obtain orthonormal set by using Gram-Schmidt Process corresponding to the set S .
3. Find an orthogonal basis for the subspace W of R^3 , given by $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x - y + 2z = 0 \right\}$.
4. Find an orthogonal basis for R^3 that contains the vector $x_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. (by Gram-Schmidt Process)
5. Identify order and degree of the following differential equations:
 - $\frac{d^2y}{dx^2} - 3x \sqrt{\frac{dy}{dx}} \frac{dy}{dx} = x.$
 - $\frac{d^2y}{dx^2} - 3x / \frac{dy}{dx} = \sin x.$
 - $y \sin \frac{dy}{dx} = x^2.$
6. Determine whether the differential equations are exact:
 - (i) $e^{x^2} (2xy \, dx + dy) = 0$
 - (ii) $(x^3 - y^2 + y \sin xy)dx - (2xy - x \sin xy)dy = 0.$If exact, solve them.
7. Find the general solution of second order linear homogeneous differential equation
$$\frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 11y = 0.$$
8. Solve the differential equation

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0, \text{ given that } y(0) = 0, y'(0) = 5. \text{ Find } y(3/2).$$

9. Find the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 2x - 3x^2.$$

10. Solve

$$\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 8y = 32e^{-2x}.$$

11. Solve

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 2x e^{-3x} \cos x.$$

12. Solve the non-homogeneous ODE by method of variation of parameter

$$\frac{d^2y}{dx^2} + y = \sec x.$$

13. Find the solution of the differential equation

$$x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0.$$

14. Solve:

$$(x^2 D^2 - xD + 1)y = \left(\frac{\log x}{x}\right)^2.$$

15. Solve the set of simultaneous differential equations

$$(3D + 1)x + 3Dy = 3t + 1$$

$$(D - 3)x + Dy = 2t.$$

16. Classify the partial differential equation

$$u_{xx} + 2(x^2 + y^2)u_{xy} + x^2 y^2 u_{yy} = 0, \text{ for region } x > 0, y > 0.$$

17. Find region in which partial differential equation

$$x^2 u_{xx} + xyu_{xy} + yu_{yy} = 0, \text{ is hyperbolic?}$$

18. Find the solution of initial value problem

$$u_t = u_{xx}, \quad u(0, t) = 0, \quad u(\pi, t) = 0 \quad \text{and} \quad u(x, 0) = \cos x \sin 5x.$$

19. Let $u(x, t)$ be the solution of initial value problem

$$u_{tt} = u_{xx}, \quad u(x, 0) = \cos 5x \quad \text{and} \quad u_t(x, 0) = 0. \quad \text{Then find value of } u(1, 1)?$$

20. Solve the Laplace equation by the separation of variables method:

$$u_{xx} + u_{yy} = 0, 0 < x < L, 0 < y < \infty$$

$$u(0, y) = u(L, y) = u(x, \infty) = 0, u(x, 0) = g(x).$$

